



**School of Computing and Engineering
National Institute of Business Management
Kandy**

**Higher National Diploma in Software Engineering
HNDSE 25.1F**

AgroSmart: -

**An IoT-Based Smart Precision Agriculture System with
Cloud Monitoring and Multi-Sensor Control**

Internet of Things – Proposal

Prepared by (Group 05) -

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Date of Submission – 11/01/2026

Declaration

We, the undersigned, hereby declare that this proposal report for AgroSmart: IoT-Based Smart Precision Agriculture System with Cloud Monitoring and Multi-Sensor Control has been prepared by us (**Group 05**) as part of a group coursework for the Internet of Things module at the National Institute of Business Management, Kandy.

We confirm that:

1. The work presented is our own and has not been submitted previously for any other academic award.
2. All sources of information, ideas, and data from other authors have been properly cited and referenced.
3. The project is carried out in accordance with the institution's academic integrity policies.

We take full responsibility for the content and authenticity of this report.

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Abstract

This project, *AgroSmart: IoT-Based Smart Precision Agriculture System with Cloud Monitoring and Multi-Sensor Control*, aims to develop a smart agriculture monitoring system integrated with a data-driven smart precision agriculture module, assisting farmers in making informed, evidence-based decisions. The system collects real-time environmental data, including temperature, humidity, soil moisture, light intensity, and water levels, using IoT sensors connected to an ESP32 microcontroller. The collected data is transmitted to a cloud platform (Firebase) and visualized through a web-based dashboard and a mobile application for real-time monitoring.

The Smart Precision Agriculture module utilizes historical environmental and crop-related datasets, along with real-time sensor data, to perform AI- and machine learning–based analysis. This enables the system to identify patterns, predict trends, and generate crop-specific insights such as growth suitability, irrigation recommendations, and seasonal climate behavior for a selected set of three crops. The project is initially focused on the Kandy region in Sri Lanka and is designed with a modular architecture to support future expansion with additional sensors, datasets, and analytical features.

The outcomes of this project demonstrate the potential of integrating IoT, cloud computing, and AI/ML techniques to support efficient resource management and sustainable agricultural practices. This project is carried out as a group coursework and provides both practical implementation experience and academic insights into modern smart farming technologies.

Acknowledgement

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Executive Summary

AgroSmart is an **IoT-based Smart Agriculture Monitoring System integrated with a Smart Precision Agriculture Module** designed to assist farmers by providing **real-time environmental monitoring and AI-based actionable insights**. The system will monitor **temperature, humidity, soil moisture, light intensity, and water levels** using a network of sensors connected to an **ESP32 microcontroller**. Data will be uploaded to the **Firebase cloud** and displayed on a **web and mobile dashboard**.

The project aims to **enhance irrigation efficiency, optimize crop selection, and forecast seasonal trends through the use of AI/ML analysis on collected sensor data**. AgroSmart will serve as a modular platform, allowing additional features and sensors to be integrated in the future.

Expected outcomes include:

- Real-time environmental data collection.
- Cloud-based visualization dashboard and mobile application.
- AI/ML-assisted recommendations for irrigation and crop selection.
- Academic documentation, including reports and presentations.

Project Background

Agriculture in regions like **Kandy, Sri Lanka**, faces multiple challenges, including inefficient water usage, unpredictable weather patterns, and reduced crop yield. Traditional farming methods **lack real-time monitoring and predictive insights**, resulting in water wastage, crop loss, and lower productivity.

With the advancement of **IoT, cloud computing, and AI**, it is now possible to monitor environmental factors in real-time, analyze data, and provide actionable recommendations. Implementing a **Smart Agriculture Monitoring System** can help farmers make **data-driven decisions**, optimize water usage, improve crop yields, and reduce resource waste.

The **motivation** behind AgroSmart is to develop an **academic-level prototype** that demonstrates the potential of **IoT and AI in precision agriculture** while providing a foundation for future enhancements.

Project Aim and Objectives

Project Aim

To design and develop an IoT-based Smart Agriculture Monitoring System integrated with a Smart Precision Agriculture module that collects real-time environmental data, analyzes historical and live data using AI and Machine Learning techniques, and provides crop-specific predictions, trends, and actionable decision-support insights for farmers.

Project Objectives

Primary Objectives

1. To design and implement an IoT-based agriculture monitoring system using multiple sensors, including:
 - Temperature & Humidity Sensor (DHT11).
 - Soil Moisture Sensors (Capacitive & Resistive).
 - Light Intensity Sensor (LDR/MH-LDR).
 - Ultrasonic Sensor for water level detection.
 - Relay-controlled pump for smart irrigation.
2. To continuously collect real-time environmental data from the deployed sensors and securely store the data on a cloud-based platform (Firebase).
3. To develop a web-based dashboard and a mobile application for real-time visualization and monitoring of environmental sensor data.
4. To collect and analyze historical environmental and crop-related datasets from external sources to understand climate patterns and crop growth behavior
5. Integrate **AI/ML algorithms** to provide:
 - Irrigation recommendations based on humidity and soil moisture.
 - Crop-specific growth suitability analysis,
 - Seasonal climate trends and predictive insights.
6. To support automated irrigation control based on analyzed sensor data and predefined decision logic
7. Ensure **modular system design** to allow future integration of additional sensors or features.

Secondary Objectives

8. To perform crop-focused analysis for a selected set of three crops by evaluating environmental conditions and growth trends.
9. To provide farmers with meaningful decision-support information rather than fully autonomous control, ensuring transparency and interpretability of results.
10. To ensure a modular and scalable system architecture that allows future integration of additional sensors, crops, analytical features, or AI models.

Scope of the Project

1. Target a specific area: **Kandy, Sri Lanka**, for initial implementation.
2. Monitor environmental parameters: **temperature, humidity, soil moisture, light intensity, and water levels.**
3. Provide actionable insights for farmers, including:
 - Optimal irrigation schedule.
 - Crop selection recommendations.
 - Predictions of climate and soil trends.
4. Modular and expandable system for additional sensors or AI modules.
5. Academic scope: real-time IoT system and AI-based recommendation system.
6. Focused on **proof-of-concept**, not large-scale industrial deployment.

Precision Agriculture Scope:

The precision agriculture module is limited to analyzing three selected crops and provides decision-support insights rather than fully autonomous control.

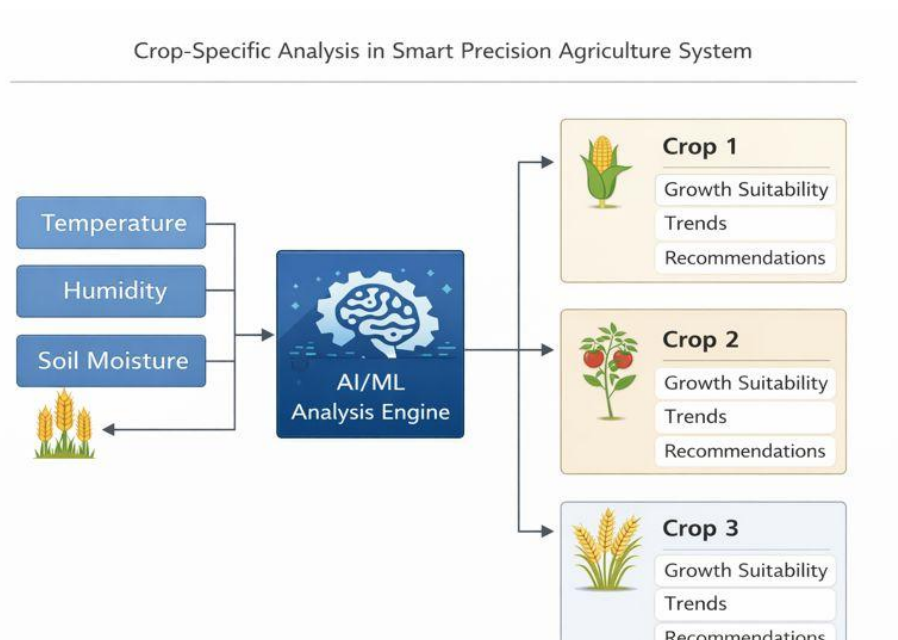


Figure 1: Precision Agriculture scope

Methodology

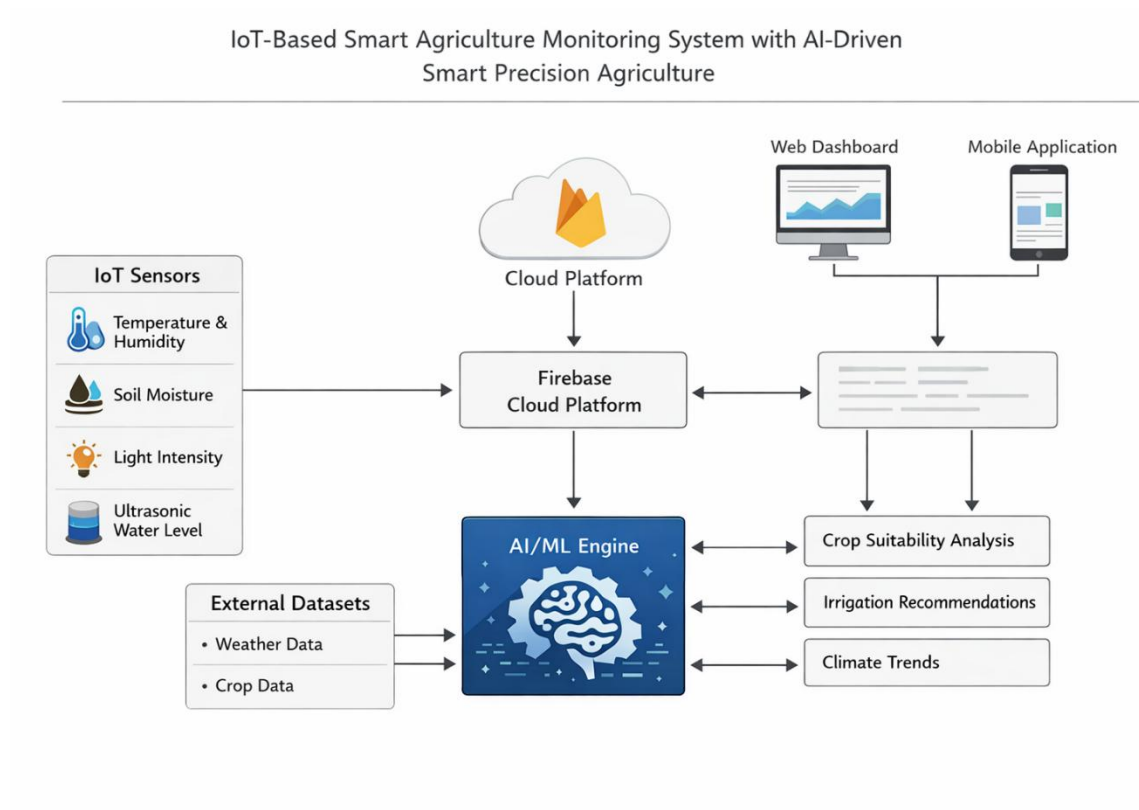


Figure 2: Overall Architecture of AgroSmart System

System Phases

1. Hardware Layer:

- ESP32 microcontroller as the main controller
- Sensors: DHT11, soil moisture, LDR, ultrasonic
- Relay module and water pump for automated irrigation
- Proper power management: 3.3V for low-voltage sensors, 5V external supply for pumps, and high-voltage sensors

2. Data Collection:

- Sensors collect real-time environmental data
- ESP32 uploads data to **Firestore Realtime Database**
- A dataset for the historical data

Sensor Name	Parameter Measured	Unit	Location	Purpose
DHT11	Temperature, Humidity	°C, %	Field environment	To monitor ambient environmental conditions affecting crop growth
Soil Moisture Sensor	Soil Moisture Level	Percentage (%)	Soil near plant root zone	To determine soil water content and enable automated irrigation decisions
Ultrasonic Water Level Sensor	Water Level	Centimeters (cm)	Water tank / reservoir	To monitor available water level for irrigation management
LDR Light Sensor	Light Intensity	Lux	Field / crop canopy area	To assess sunlight availability for photosynthesis analysis

Figure 3: Sensor Data Collected by AgroSmart

3. Data Analysis & AI/ML Module:

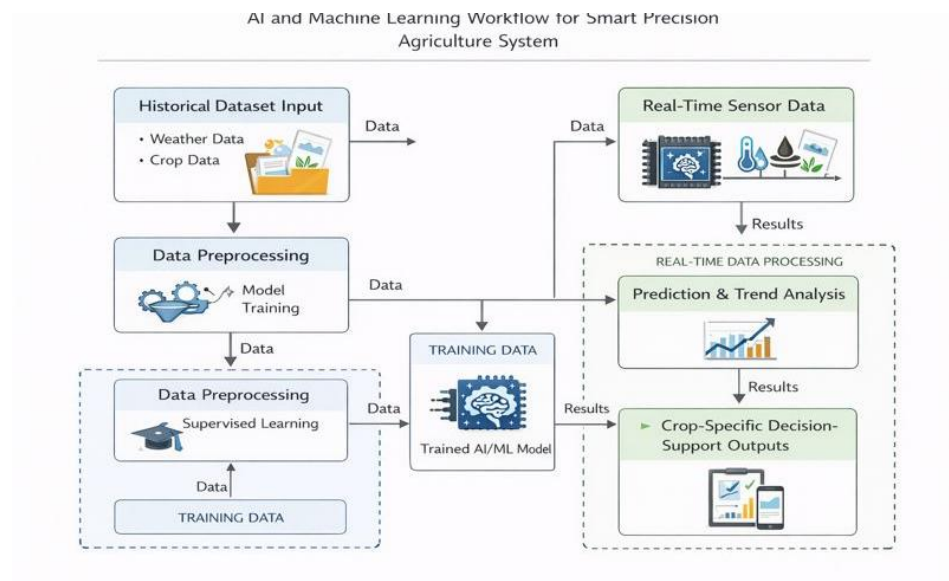


Figure 4: Data Flow from Sensors to Cloud and AI Engine

- Use of external datasets (Kaggle, public sources)
- Learning environmental patterns
- Crop-specific analysis
- Decision-support outputs

- AI/ML approach:
 - Supervised learning for historically labeled data is available
 - Unsupervised learning for pattern detection in unlabeled data

4. Data Description:

- **Dataset data** (historical weather, crop data)
- **Real-time sensor data**
- Data types:
 - Numerical
 - Time-series
 - Labeled

5. Dashboard and Mobile Application:

- Visualize sensor readings in real-time
- Display recommendations and alerts
- Allow remote monitoring of irrigation status

6. Testing & Validation:

- Test sensors individually and in combination
- Calibrate soil moisture thresholds and water tank levels
- Validate AI predictions using sample datasets

7. AI-Based Smart Precision System

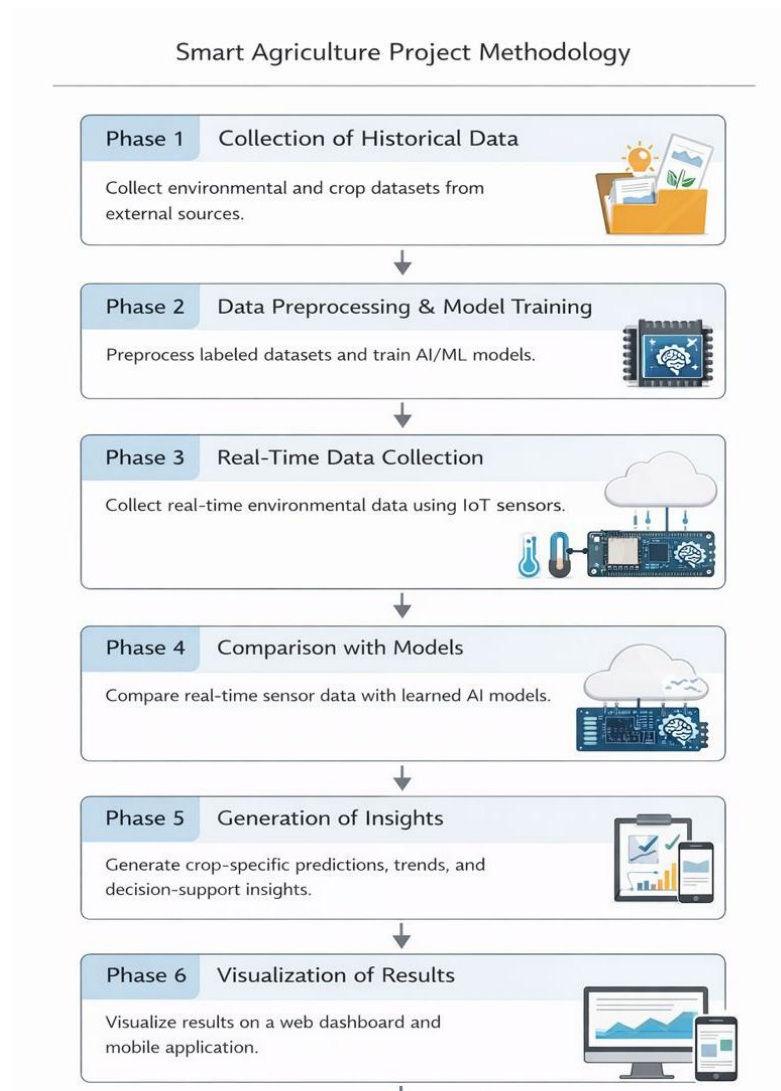


Figure 5: AI and Data Analysis Flow

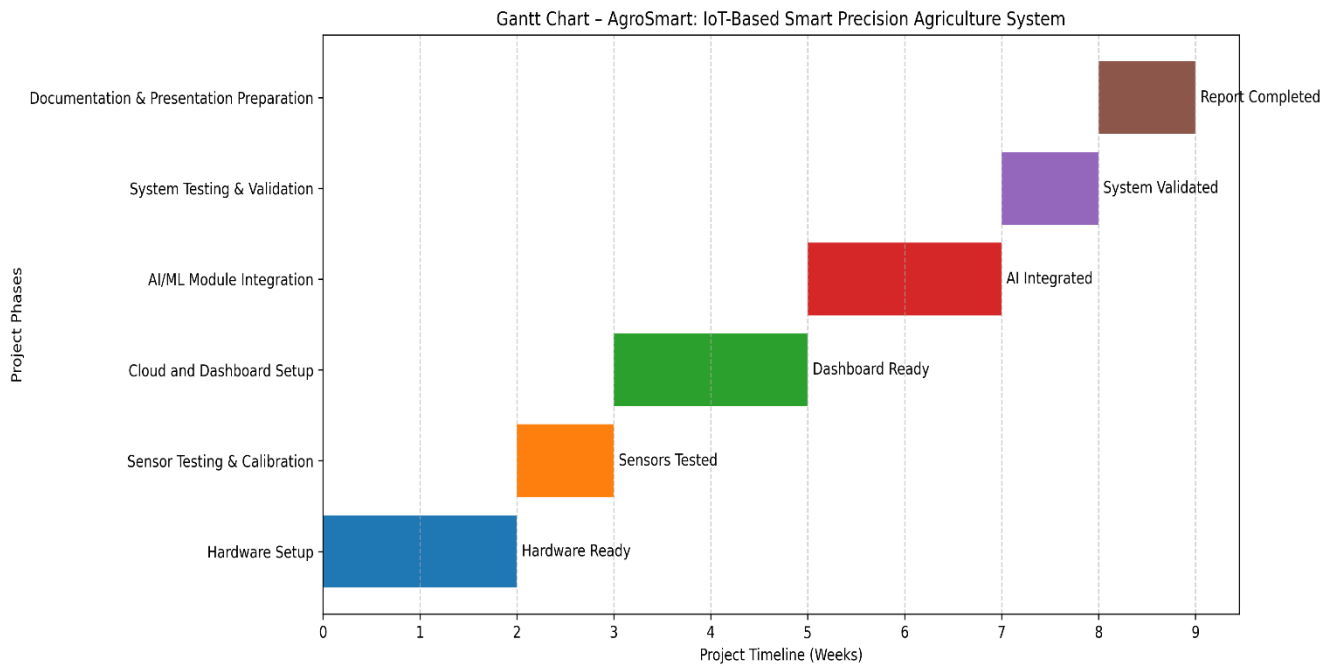
Tools and Technologies

- Microcontroller: ESP32
- Sensors: DHT11, Capacitive/Normal Soil Moisture, LDR, Ultrasonic
- Cloud Platform: Firebase Realtime Database
- AI/ML: Python, TensorFlow / scikit-learn
- Dashboard / App: Web technologies (HTML/CSS/JS) or Flutter for mobile
- Version Control & collaboration: GitHub - Cloud-based Version Control System

Project Deliverables

1. **Hardware Prototype:** ESP32 + sensor network + relay/pump system
2. **Software Module:** Cloud-based data storage, web dashboard, mobile app
3. **AI/ML Module:**
 - Crop suitability analysis
 - Trend prediction outputs
 - AI-generated insights visualized on the dashboard
4. **Reports & Documentation:** Proposal report, final report
5. **Presentations:** Proposal, progress updates, final presentation

Timeline



Phase	Duration
Hardware setup	2 weeks
Sensor testing & calibration	1 week
Cloud and dashboard setup	2 weeks
AI/ML module integration	2 weeks
System testing & validation	1 week
Documentation & presentation preparation	1 week

References

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- Espressif Systems. (2026). *ESP32 Documentation*. <https://www.espressif.com>
- Zhang, L., & Kumar, A. (2020). *AI-driven smart irrigation systems: A review*. Agricultural Engineering Journal, 15(2), 110–125.

Appendices

- **Hardware Block Diagram:** ESP32 connected to sensors, relay, and pump
- **System Architecture Diagram:** Data flow from sensors → ESP32 → Firebase → AI module → Dashboard
- **Sample Sensor Data Table:** Example readings for temperature, humidity, soil moisture, and light intensity
- **Additional Figures/Charts:** Sensor calibration charts, optional images of sensors